Theorem 5.1.1:

$$\frac{\prod_{1\le i<j\le \ell}(1-qt\,x_i y_j)}{\prod_{1\le i\le j\le \ell}(1-t\,x_i y_j)} = \sum_{\vec{a}\in\mathbb{N}^\ell} t^{|\vec{a}|} E^\sigma_{\vec{a}}(x_1,\dots,x_\ell;q^{-1})\,F^\sigma_{\vec{a}}(y_1,\dots,y_\ell;q) \qquad \forall \sigma\in S_\ell.$$

$|\vec{a}| = a_1+a_2+\dots+a_\ell$

$\vec{a}=(a_1,\dots,a_\ell),\ a_1,a_2,\dots,a_\ell\ge 0$

Proof: We first show that for any $f(y)\in\mathbb{Q}(q)[y_1,\dots,y_\ell]$,

$$f(tx) = \langle y^0\rangle f(y)\frac{\prod_{1\le i<j\le\ell}\left(1-q^{-1}t\frac{x_i}{y_j}\right)}{\prod_{1\le i\le j\le\ell}\left(1-t\frac{x_i}{y_j}\right)}\prod_{i<j}\frac{1-\frac{y_i}{y_j}}{1-q^{-1}\frac{y_i}{y_j}}$$

Note that the terms involving y_1 are

$$\frac{1}{1-t\frac{x_1}{y_1}}\cdot\prod_{1<j\le\ell}\frac{1-\frac{y_1}{y_j}}{1-q^{-1}\frac{y_1}{y_j}} = \frac{1}{1-t\frac{x_1}{y_1}}\left(1-\frac{y_1}{y_2}\right)\left(1-\frac{y_1}{y_3}\right)\cdots\left(1-\frac{y_1}{y_\ell}\right)\left(1+q^{-1}\frac{y_1}{y_2}+q^{-2}\frac{y_1^2}{y_2^2}+\cdots\right)\cdots\left(1+q^{-1}\frac{y_1}{y_\ell}+q^{-2}\frac{y_1^2}{y_\ell^2}+\cdots\right)$$

For any power series $g(y_1)$, we have $\langle y_1^0\rangle g(y_1)\frac{1}{1-\frac{tx_1}{y_1}} = g(tx_1)$

Let $g(y_1)=\sum_{i\ge 0}a_i y_1^i$.

Then $g(y_1)\cdot\frac{1}{1-\frac{tx_1}{y_1}} = \sum_{i\ge 0}a_i y_1^i\cdot\sum_{j\ge 0}\frac{(tx_1)^j}{y_1^j} = \sum_{i,j\ge 0}a_i(tx_1)^j y_1^{i-j}$

$\therefore\ \langle y_1^0\rangle g(y_1)\cdot\frac{1}{1-\frac{tx_1}{y_1}} = \sum_{i\ge 0}a_i(tx_1)^i$ (when $j=i$)

$= g(tx_1)$

Next, note that

$$\frac{\prod_{1\le i<j\le\ell}\left(1-q^{-1}t\frac{x_i}{y_j}\right)}{\prod_{1\le i\le j\le\ell}\left(1-t\frac{x_i}{y_j}\right)}\prod_{i<j}\frac{1-\frac{y_i}{y_j}}{1-q^{-1}\frac{y_i}{y_j}} = \frac{1}{1-\frac{tx_1}{y_1}}\cdot\underbrace{\boxed{\frac{\prod_{1<i<j\le\ell}\left(1-q^{-1}t\frac{x_i}{y_j}\right)}{\prod_{1<j\le\ell}\left(1-t\frac{x_1}{y_j}\right)}\cdot\prod_{1<j\le\ell}\frac{1-\frac{y_1}{y_j}}{1-q^{-1}\frac{y_1}{y_j}}}}_{=\,1 \text{ when } y_1=tx_1}\cdot\frac{\prod_{2\le i<j\le\ell}\left(1-q^{-1}t\frac{x_i}{y_j}\right)}{\prod_{2\le i\le j\le\ell}\left(1-t\frac{x_i}{y_j}\right)}\prod_{2\le i<j\le\ell}\frac{1-\frac{y_i}{y_j}}{1-q^{-1}\frac{y_i}{y_j}}$$

Hence

$$\langle y_1^0\rangle f(y)\frac{\prod_{1\le i<j\le\ell}\left(1-q^{-1}t\frac{x_i}{y_j}\right)}{\prod_{1\le i\le j\le\ell}\left(1-t\frac{x_i}{y_j}\right)}\prod_{i<j}\frac{1-\frac{y_i}{y_j}}{1-q^{-1}\frac{y_i}{y_j}} = \langle y_1^0\rangle f(y)\frac{\prod_{1<i<j\le\ell}\left(1-q^{-1}t\frac{x_i}{y_j}\right)}{\prod_{1<j\le\ell}\left(1-t\frac{x_i}{y_j}\right)}\prod_{1<j\le\ell}\frac{1-\frac{y_1}{y_j}}{1-q^{-1}\frac{y_1}{y_j}}\cdot\frac{\prod_{2\le i<j\le\ell}\left(1-q^{-1}t\frac{x_i}{y_j}\right)}{\prod_{2\le i\le j\le\ell}\left(1-t\frac{x_i}{y_j}\right)}\prod_{2\le i<j\le\ell}\frac{1-\frac{y_i}{y_j}}{1-q^{-1}\frac{y_i}{y_j}}\Bigg|_{y_1=tx_1}$$

$$= f(tx_1,y_2,\dots,y_\ell)\cdot\frac{\prod_{2\le i<j\le\ell}\left(1-q^{-1}t\frac{x_i}{y_j}\right)}{\prod_{2\le i\le j\le\ell}\left(1-t\frac{x_i}{y_j}\right)}\prod_{2\le i<j\le\ell}\frac{1-\frac{y_i}{y_j}}{1-q^{-1}\frac{y_i}{y_j}}$$

Similarly,

$$\langle y_1^0 y_2^0\rangle f(y)\frac{\prod_{1\le i<j\le\ell}\left(1-q^{-1}t\frac{x_i}{y_j}\right)}{\prod_{1\le i\le j\le\ell}\left(1-t\frac{x_i}{y_j}\right)}\prod_{i<j}\frac{1-\frac{y_i}{y_j}}{1-q^{-1}\frac{y_i}{y_j}} = f(tx_1,tx_2,y_3,\dots,y_\ell)\cdot\frac{\prod_{3\le i<j\le\ell}\left(1-q^{-1}t\frac{x_i}{y_j}\right)}{\prod_{3\le i\le j\le\ell}\left(1-t\frac{x_i}{y_j}\right)}\prod_{3\le i<j\le\ell}\frac{1-\frac{y_i}{y_j}}{1-q^{-1}\frac{y_i}{y_j}}$$

$\vdots$

$$\therefore\ \langle y^0\rangle f(y)\frac{\prod_{1\le i<j\le\ell}\left(1-q^{-1}t\frac{x_i}{y_j}\right)}{\prod_{1\le i\le j\le\ell}\left(1-t\frac{x_i}{y_j}\right)}\prod_{i<j}\frac{1-\frac{y_i}{y_j}}{1-q^{-1}\frac{y_i}{y_j}} = f(tx_1,\dots,tx_\ell) = f(tx) \text{ which proves the first claim.}$$

By definition, $E^\sigma_{\vec{a}}(x_1,\dots,x_\ell;q),\ F^\sigma_{\vec{a}}(x_1,\dots,x_\ell;q)\in\mathbb{Q}(q)[x_1,\dots,x_\ell]$.

$\{E^\sigma_{\vec{a}}:\vec{a}\in\mathbb{N}^\ell\}$ forms a graded basis for $\mathbb{Q}(q)[x_1,\dots,x_\ell]$ because they are homogeneous with leading term $x^{\vec{a}}$.
Similarly, $\{F^\sigma_{\vec{a}}:\vec{a}\in\mathbb{N}^\ell\}$ also forms a graded basis for $\mathbb{Q}(q)[x_1,\dots,x_\ell]$.

Let $Z(x,y,q,t) = \frac{\prod_{1\le i<j\le\ell}(1-qt\,x_i y_j)}{\prod_{1\le i\le j\le\ell}(1-t\,x_i y_j)}$ and put $f(y)=E^\sigma_{\vec{a}}(y_1,\dots,y_\ell;q)$

$$\therefore\ E^\sigma_{\vec{a}}(tx_1,tx_2,\dots,tx_\ell;q) = \langle y^0\rangle E^\sigma_{\vec{a}}(y_1,\dots,y_\ell;q)\,Z(x,y^{-1},q^{-1},t)\prod_{i<j}\frac{1-\frac{y_i}{y_j}}{1-q^{-1}\frac{y_i}{y_j}} = \left\langle E^\sigma_{\vec{a}}(y_1,\dots,y_\ell;q),\ Z(x,y^{-1},q^{-1},t)\right\rangle_q$$

By Prop. 4.3.2, $\{E^\sigma_\lambda(y;q)\}_\lambda$ and $\{\overline{F^\sigma_\lambda(y;q)}\}_\lambda$ are dual bases w.r.t. $\langle\ ,\ \rangle_q$

Hence coeff of $\overline{F^{\sigma}_{\vec{a}}(y;q)}$ in $Z(x,y^{-1},q^{-1},t) = E^{\sigma}_{\vec{a}}(tx_1, tx_2, \dots, tx_\ell; q) = t^{|\vec{a}|} E^{\sigma}_{\vec{a}}(x;q)$

$\therefore$ coeff of $F^{\sigma}_{\vec{a}}(y;q)$ in $Z(x,y,q,t) = t^{|\vec{a}|} E^{\sigma}_{\vec{a}}(x;q^{-1})$

Hence $Z(x,y,q,t) = \sum_{\vec{a}\in\mathbb{N}^\ell} t^{|\vec{a}|} E^{\sigma}_{\vec{a}}(x_1,\dots,x_\ell;q^{-1}) F^{\sigma}_{\vec{a}}(y_1,\dots,y_\ell;q)$.

$\square$